Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	31 January 2026
Team ID	[Enter Team ID]
Project Name	Global Fuel Prices and Taxation Analysis (2020 to 2026)
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the technologies, platforms, databases, BI tools, and integrations that you will use to build your Business Intelligence solution.

S.No	Component	Description	Technology	Justification / Purpose
1.	Programming / Query Language	Language used to query, filter, transform, aggregate, and analyze data.	Python (pandas), Tableau calculated fields, HTML, CSS, JavaScript	Python checks and summarizes the data; Tableau calculations create subsidy score, average price and volatility; HTML, CSS and JavaScript build the web dashboard.
2.	Database / Data Warehouse	Warehouse used to store and manage structured business data for analysis and reporting.	CSV (global_fuel_prices_Dataset.csv)	A single flat file of 27,468 records is small enough that a warehouse is not needed, and Tableau reads it directly.
3.	Data Ingestion	Used to load data from source files or external systems into the analytical environment.	Tableau Data Connections (Text file)	Loads the CSV into Tableau with no extra setup and refreshes when the file changes.
4.	Data Transformation and Business Logic	Tools/Logic used to clean, join, filter, aggregate data for analysis and calculate KPIs, metrics, business rules, and analytical measures.	Tableau Desktop	Sets data types, geographic roles and calculated fields (Subsidy Score, Avg Fuel Price, Diesel Volatility) and applies filters and aggregations.
5.	Data Visualization / Analytics	Used to create interactive dashboards, charts, reports, KPIs, and visual analysis.	Tableau Desktop, Python	Builds the map, scatter, forecast, ranking, heatmap and income charts; Python supports checks and the web version.
6.	Front-End / User Interface	Interface through which users interact with dashboards, filters, parameters, and visual insights.	Tableau dashboard (Tableau Public), HTML web dashboard	Gives users filters, click-to-filter and tooltips in a browser without installing software.